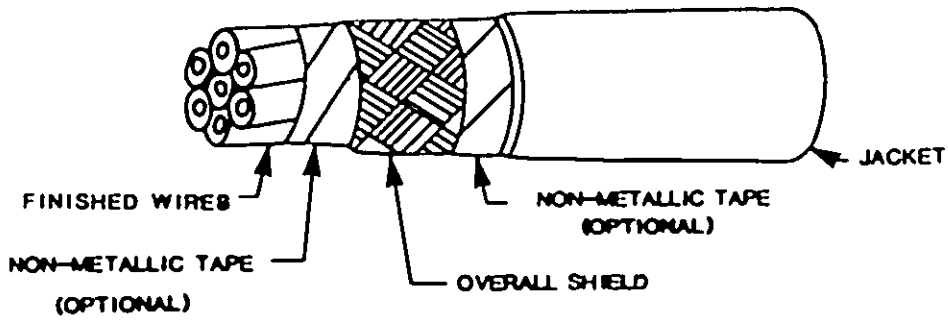
MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-CONDUCTOR, AWG 16, TYPE MXSO

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24640.



SH 12757

FINISHED WIRE: White MIL-W-81044/12-16, except identification codes and methods shall be in accordance with MIL-C-24640.

FIGURE 1. Construction (non-watertight).

FSC 6145

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TABLE III. Construction details.

Type number	Number of conductors	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)
MXSO-2	2	0.175	0.034	0.255	0.275	57.4
MXSO-9	9	.325	.041	.415	.447	160.0
MXSO-21	21	.456	.044	.549	.591	316.0
MXSO-37	37	.562	.044	.651	.701	488.0

REQUIREMENTS:

Qualification required.
Construction details: See table I.

Overall shield shall be tin-coated copper braid with a coverage not less than 85 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	4.90
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2500
Conductor to shield	1000
Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	Not applicable
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	.35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor - conformance to material, construction and coverage	Not applicable
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - ME, AR
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0903-10)

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